

Scope of Project:

Company: Mystic Falls Community Digital Library system

Introduction:

The purpose of this project is to design & implement a database system for the Mystic Falls Community Digital library system. This system supports digital access to reading materials by allowing members to browse, borrow, and reserve eBooks.

The system allows users to browse, reserve, and borrow eBooks through a structured lending process similar to modern digital library applications such as Libby. Each loan records a borrowing track record for tracking access to eBooks.

The system manages library operations including eBook availability, member borrowing activity, and reservation tracking. It ensures that all transactions are properly recorded and that data consistency is maintained across all users and resources.

Data requirements:

The system will store data related to eBooks including book title, book author, book id, description, availability status, publisher and format (PDF, EPUB, AudioBook) information. The eBooks will also be associated with certain categories to help with users searching for books.

Member data will include: name, email, member ID and account status. It will also track member borrowing activity.

Loan data records the borrowing transactions, which would include: loanID, memberID, eBook ID, loan date, return date.

The reservation data would track users requests for eBooks that are not available yet. This would include: reservation ID, memberID, ebook ID, reservation date and reservation status (hold/ready) .

Each eBook is associated with a library branch to support branch based organization and availability.

eBooks will also be associated with an author, publisher, and category through relationships in the database.

Business Processes:

The eBook borrowing process allows members to borrow available eBooks. A loan record is created for each transaction and the system is updated to reflect the loan status.

The eBook return process updates the loan record with a return date and makes the eBook available again.

The reservation process allows members to place holds on unavailable eBooks. Reservations are stored and fulfilled when the eBook becomes available.

The catalog management process allows librarians to manage eBooks and related data such as authors, publishers, and categories.

The member management process maintains member account information. Only active members can borrow books.

User Needs:

1. **Members-** They need to be able to borrow eBooks, browse available books, return ebooks, place holds on unavailable books, view status/accounts
2. **Librarians-** They need to be able to manage catalogs (add, update, remove ebooks), monitor loans/reservations, manage member accounts and ensuring ebook availability status is accurate

Entities:

- eBook
- members
- loans
- reservations
- librarian
- branch
- format
- author
- publisher
- categories

Relationships: We can identify the following relationships between the entities:

- **eBook & loans (1:M)**(A single loan record is created for exactly **one eBook**, a single **eBook** can have **many loan records** over time.)
- **reservations & members (1:M)** (A single reservation belongs to **one member(1:1)**, a single **member** is permitted to hold **many eBook reservations** at a time.)
- **reservations & eBooks (1:M)** (A single reservation is associated with **one eBook-1:1**, a single **eBook** can have **many active reservations.**)
- **loans & members (1:M)** A single loan allows **one member** to borrow or extend an item, a single **member** can be associated with **many loan transactions/extensions** over time)
- **librarians & eBooks (1:M)** (One librarian manages many eBooks, each **eBook** is managed by exactly **one librarian.1:1**)
- **author & eBook (1:M)** (Each **author** can have **many eBooks**, each individual **eBook** entry tracks back to exactly **one author -1:1**).
- **publisher & eBook (1:M)** (Each **publisher** can distribute **many eBooks**, each individual **eBook** is published by exactly **one publisher. -1:1**)
- **categories& eBook (1:M)** (Each category can be associated to **many eBooks**, 1 eBook is under **1 category 1:1**)
- **eBook & format (1:M)** (Each eBook has 1 format -1:1, 1 format can be used for many eBooks-1:M)
- **branch & eBook (1:M)** (1 branch can manage **many ebooks -1:M**, 1 eBook belongs to **1 branch**)